PROGRAM:

AIES-Exp_3.py - C:/Users/ABBIRAMY/OneDrive/Documents/AIES_LAB/AIES-Exp_3.py (3.14.2)

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```
tree = [[[3, 5], [6, 9]], [[1, 2], [0, -1]]]
leaves_visited = 0
def alphabeta(node, is_max, alpha, beta):
    global leaves_visited
    if not isinstance(node, list):
        leaves_visited += 1
        return node
    if is_max:
        best = float('-inf')
        for child in node:
            best = max(best, alphabeta(child, False, alpha, beta))
            alpha = max(alpha, best)
            if beta <= alpha:
                break
        return best
    else:
        best = float('inf')
        for child in node:
            best = min(best, alphabeta(child, True, alpha, beta))
            beta = min(beta, best)
            if beta <= alpha:
                break
        return best
value = alphabeta(tree, True, float('-inf'), float('inf'))
print("Best value for MAX:", value)
print("Leaves visited    :", leaves_visited, "out of 8")
```

OUTPUT:

IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/ABBIRAMY/OneDrive/Documents/AIES_LAB/AIES-Exp_3.py =====
Best value for MAX: 5
Leaves visited    : 5 out of 8
>>>
```